

Some other Numericals on Set Theory

15

Q → For any sets A, B and C prove the following mathematically :

$$(a) (A-B) \cup (B-A) = (A \cup B) - (A \cap B)$$

$$(b) (A-B) - C = A - (B \cup C)$$

$$(c) (A-B) - C = (A-C) - B$$

$$(d) (A-B) - C = (A-C) - (B-C)$$

$$(e) A \subseteq A \cup B \text{ and } A \cap B \subseteq A$$

Solⁿ →

$$(a) \text{ Let } x \in (A-B) \cup (B-A)$$

$$\Rightarrow x \in (A-B) \text{ or } x \in (B-A)$$

$$\Rightarrow (x \in A \text{ and } x \notin B) \text{ or } (x \in B \text{ and } x \notin A)$$

$$\Rightarrow (x \in A \text{ or } x \in B) \text{ and } (x \notin B \text{ or } x \notin A)$$

$$\Rightarrow x \in A \cup B \text{ and } x \notin A \cap B$$

$$\Rightarrow x \in (A \cup B) - (A \cap B)$$

$$\text{Hence } (A-B) \cup (B-A) \subseteq (A \cup B) - (A \cap B) \rightarrow (1)$$

Again

$$x \in (A \cup B) - (A \cap B)$$

$$\Rightarrow x \in A \cup B \quad \text{and} \quad x \notin A \cap B$$

$$\Rightarrow (x \in A \text{ or } x \in B) \quad \text{and} \quad (x \notin A \text{ or } x \notin B)$$

$$\Rightarrow (x \in A \text{ and } x \notin B) \quad \text{or} \quad (x \in B \text{ and } x \notin A)$$

$$\Rightarrow x \in (A-B) \text{ or } x \in (B-A)$$

$$\Rightarrow x \in (A-B) \cup (B-A)$$

$$\therefore (A \cup B) - (A \cap B) = (A-B) \cup (B-A) \rightarrow \textcircled{2}$$

From ① & ②

$$(A \cup B) - (A \cap B) = (A-B) \cup (B-A)$$

$$(b) \quad x \in (A-B) - C$$

$$\Rightarrow x \in (A-B) \text{ and } x \notin C$$

$$\Rightarrow (x \in A \text{ and } x \notin B) \text{ and } x \notin C$$

$$\Rightarrow x \in A \text{ and } (x \notin B \text{ and } x \notin C)$$

$$\Rightarrow x \in A \text{ and } (x \notin B \cup C)$$

$$\Rightarrow x \in A - (B \cup C)$$

$$\text{Hence } (A-B) - C \subseteq A - (B \cup C)$$

Again Let $x \in A - (B \cup C)$

$$\Rightarrow x \in A \text{ and } x \notin (B \cup C)$$

$$\Rightarrow x \in A \text{ and } (x \notin B \text{ and } x \notin C)$$

$$\Rightarrow (x \in A \text{ and } x \notin B) \text{ and } x \notin C$$

$$\Rightarrow x \in (A - B) \text{ and } x \notin C$$

$$\Rightarrow x \in (A - B) - C$$

$$\text{Hence } A - (B \cup C) \subseteq (A - B) - C \longrightarrow \textcircled{2}$$

From ① & ② we have,

$$\underline{(A - B) - C = A - (B \cup C)}$$

(C) Let $x \in (A - B) - C$.

$$\Rightarrow x \in (A - B) \text{ and } x \notin C$$

$$\Rightarrow (x \in A \text{ and } x \notin B) \text{ and } x \notin C$$

$$\Rightarrow (x \in A \text{ and } x \notin C) \text{ and } x \notin B$$

$$\Rightarrow x \in (A - C) \text{ and } x \notin B$$

$$\Rightarrow x \in (A - C) - B$$

$$\text{Hence } (A - B) - C \subseteq (A - C) - B \longrightarrow \textcircled{1}$$

Again Let

$$x \in (A - C) - B$$

$$\Rightarrow x \in (A - C) \text{ and } x \notin B$$

$$\Rightarrow (x \in A \text{ and } x \notin C) \text{ and } x \notin B$$

$$\Rightarrow (x \in A \text{ and } x \notin B) \text{ and } x \notin C$$

$$\Rightarrow (x \in (A - B)) \text{ and } x \notin C$$

$$\Rightarrow x \in (A - B) - C$$

Hence

$$(A - C) - B \subseteq (A - B) - C \rightarrow \textcircled{2}$$

From ① & ② we have

$$(A - B) - C = (A - C) - B$$

(d) Let $x \in (A - B) - C$

$$\Rightarrow x \in (A - B) \text{ and } x \notin C$$

$$\Rightarrow (x \in A \text{ and } x \notin B) \text{ and } x \notin C$$

$$\Rightarrow (x \in A \text{ and } x \notin C) \text{ and } x \notin B$$

$$\Rightarrow x \in (A - C) \text{ and } x \notin (B - C)$$

$$\therefore B - C \subseteq B$$

$$x \in (A-C) - (B-C)$$

(17)

Hence

$$(A-B) - C \subseteq (A-C) - (B-C) \rightarrow (1)$$

Again Let

$$x \in (A-C) - (B-C)$$

$$\Rightarrow x \in (A-C) \text{ and } x \notin (B-C)$$

$$\Rightarrow (x \in A \text{ and } x \notin C) \text{ and } (x \notin B \text{ and } x \notin C)$$

$$\Rightarrow (x \in A \text{ and } x \notin C) \text{ and } (x \notin B)$$

$$\Rightarrow (x \in A \text{ and } x \notin B) \text{ and } x \notin C$$

$$\Rightarrow x \in (A-B) \text{ and } x \notin C$$

$$\Rightarrow x \in (A-B) - C$$

$$\Rightarrow (A-C) - (B-C) \subseteq (A-B) - C \rightarrow (2)$$

Now from (1) and (2)

$$(A-B) - C = (A-C) - (B-C)$$

(c) Let $x \in A$

This implies that x will always lie in the set which results from the union of any other set with A .

$$\Rightarrow x \in A \cup B$$

$$\text{Hence } A \subseteq A \cup B \quad \longrightarrow \textcircled{1}$$

$$\textcircled{*} \text{ Again let } x \in A \cap B$$

$$\Rightarrow x \in A \text{ and } x \in B$$

This means that if x lies in $(A \cap B)$ then x will always lie in the set A .

$$\Rightarrow x \in A$$

$$\text{Thus } A \cap B \subseteq A \quad \longrightarrow \textcircled{2}$$

Ex $\rightarrow$ Prove the following. (do not use examples)

$$\textcircled{*} \text{ (i) } \overline{A \cup (B \cap C)} = (\bar{C} \cup \bar{B}) \cap \bar{A}$$

$$\text{(ii) } A \oplus B = (A \cup B) - (A \cap B)$$

$$\text{Soln} \rightarrow \text{(i) } \overline{A \cup (B \cap C)} = (\bar{C} \cup \bar{B}) \cap \bar{A}$$

$$\text{Let } x \in \overline{A \cup (B \cap C)} \Rightarrow x \notin A \cup (B \cap C)$$

$$\Rightarrow (x \notin A) \text{ and } (x \notin B \cap C)$$

$$\Rightarrow (x \notin A) \text{ and } (x \notin B \text{ or } x \notin C)$$

$$\Rightarrow (x \in \bar{A}) \text{ and } (x \in \bar{B} \text{ or } x \in \bar{C})$$

$$x \in \bar{A} \text{ and } x \in (\bar{C} \cup \bar{B})$$

(18)

$$\Rightarrow x \in \bar{A} \text{ and } x \in (\bar{C} \cup \bar{B})$$

$$\Rightarrow x \in (\bar{C} \cup \bar{B}) \text{ and } x \in (\bar{A})$$

$$\Rightarrow x \in (\bar{C} \cup \bar{B}) \cap \bar{A}$$

$$\therefore \overline{A \cup (B \cap C)} \subseteq (\bar{C} \cup \bar{B}) \cap \bar{A} \rightarrow (1)$$

Again let

$$x \in (\bar{C} \cup \bar{B}) \cap \bar{A}$$

$$\Rightarrow x \in (\bar{C} \cup \bar{B}) \text{ and } x \in \bar{A}$$

$$\Rightarrow \cancel{x \in \bar{B}} \Rightarrow (x \in \bar{C} \text{ or } x \in \bar{B}) \text{ and } x \notin A$$

$$\Rightarrow x \notin A \text{ and } (x \notin C \text{ or } x \notin B)$$

$$\Rightarrow x \notin A \cup (B \cap C)$$

$$\Rightarrow x \in \overline{A \cup (B \cap C)}$$

$$\therefore (\bar{C} \cup \bar{B}) \cap \bar{A} \subseteq \overline{A \cup (B \cap C)} \rightarrow (2)$$

From (1) & (2) we have

$$\overline{A \cup (B \cap C)} = (\bar{C} \cup \bar{B}) \cap \bar{A}$$

$$(ii) \quad A \oplus B = (A - B) \cup (B - A) = (A \cup B) - (A \cap B)$$

We know that

$$A \oplus B = (A - B) \cup (B - A)$$

$$\text{Let } x \in A \oplus B \Rightarrow x \in (A - B) \cup (B - A)$$

$$\Rightarrow x \in (A - B) \text{ or } x \in (B - A)$$

$$\Rightarrow (x \in A \text{ and } x \notin B) \text{ or } (x \in B \text{ and } x \notin A)$$

$$\Rightarrow (x \in A \text{ or } x \in B) \text{ and } (x \notin A \text{ or } x \notin B)$$

$$\Rightarrow x \in A \cup B \text{ and } x \notin A \cap B$$

$$\Rightarrow x \in (A \cup B) - (A \cap B)$$

$$\therefore A \oplus B \subseteq (A \cup B) - (A \cap B) \longrightarrow \textcircled{1}$$

Again let

$$x \in (A \cup B) - (A \cap B)$$

$$\Rightarrow x \in (A \cup B) \text{ and } x \notin A \cap B$$

$$\Rightarrow \cancel{x \in A \text{ and } x \notin B}$$

$$\Rightarrow (x \in A \text{ or } x \in B) \text{ and } (x \notin A \text{ or } x \notin B)$$

$$\Rightarrow (x \in A \text{ and } x \notin B) \text{ or } (x \in B \text{ and } x \notin A)$$

$$\Rightarrow (x \in A - B) \text{ or } x \in (B - A)$$

$$\Rightarrow x \in (A - B) \cup (B - A) \Rightarrow x \in A \oplus B$$

$$\therefore (A \cup B) - (A \cap B) \subseteq A \oplus B \longrightarrow (2)$$

From (1) & (2) we get

$$A \oplus B = (A \cup B) - (A \cap B)$$

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